A IMPLEMENTATION DETAILS

A.1 Data Preparation

We split all datasets using a ratio of 7:2:1 for training, validation, and testing sets, respectively. We used an input sequence length of 96 and an output sequence length of 96 to be consistent with prior work. For datasets with shorter temporal extents (e.g., Milano-6165 and ILI), we used input and output lengths of 36, following the alternative setting used in previous studies.

A.2 Training and Evaluation Configuration

For training the compared models, we set the maximum training epochs to 30 and applied early stopping after 10 epochs without validation improvement. We conducted five independent runs with different random seeds and reported the average values. All experiments were conducted on a server running Ubuntu 22.04.5 LTS with 14 CPU cores allocated from an Intel Xeon 6960P processor (72 cores total), 2TB of RAM, and 23GB of GPU memory allocated from an NVIDIA B200 (192GB total). We evaluate model performance using Mean Squared Error (MSE), inference time, and peak GPU memory.

A.3 Hyperparameter Optimization

We employ the Optuna framework [2], utilizing Bayesian optimization with TPESampler for systematic hyperparameter exploration. Each model is tuned with model-specific hyperparameter pools, with detailed configurations provided in the code repository, to ensure optimal performance under both CD and CI conditions. The optimization process aims to minimize MSE on the validation set. For each model-dataset pair, we evaluate 10 candidate configurations and select the best-performing one for final training and evaluation.

B FORMAL DEFINITIONS OF THE THREE AXES

This section provides the full formulations that Section 4 summarizes with one equation per category.

B.1 Interaction Architecture Classification

The interaction architecture dimension adopts the conventional backbone-based classification approach, organizing models into four primary architectural categories introduced in Section 2.2. This dimension serves as the foundational classification layer while remaining compatible with our extended framework. We outline the theoretical definitions and core components of the four base architectures that serve as foundational building blocks for the compared MTSF models, as illustrated in Figure 2.

MLP-based Models. These models utilize Multi-Layer Perceptrons as their primary computational component, employing linear layers to project time series data across temporal and channel dimensions. The fundamental operation can be expressed as:

$$\mathbf{Y} = \mathbf{W}_2\sigma(\mathbf{W}_1\mathbf{X} + \mathbf{b}_1) + \mathbf{b}_2, \tag{14}$$

where $\mathbf{X} \in \mathbb{R}^{T\times C}$ represents the input time series, $\mathbf{W}_1, \mathbf{W}_2$ are weight matrices, and σ denotes the activation function. Representative models include Linear and DLinear, which have demonstrated competitive performance despite their architectural simplicity.

RNN-based Models: Recurrent Neural Networks and their variants process sequential dependencies through temporal state propagation. These models maintain hidden states that encode temporal information across all channels simultaneously:

$$h_t = W_h X_t + W_{hh} h_{t-1} + b_h, \tag{15}$$

$$Y_t = W_{out} h_t + b_{out}, \tag{16}$$

where $X_t \in \mathbb{R}^C$ represents all channels at time t, $W_h \in \mathbb{R}^{H\times C}$ is the input-to-hidden weight matrix, $W_{hh} \in \mathbb{R}^{H\times H}$ is the hidden-to-hidden weight matrix, $h_t \in \mathbb{R}^H$ captures temporal dependencies across channels, where W_{out} and b_{out} denote the output projection weights and bias. Examples include vanilla RNN and SegRNN.

CNN-based Models. Convolutional Neural Networks apply convolution operations to extract localized patterns from both temporal and channel dimensions. The multi-dimensional convolution operation is defined as:

$$\mathbf{Y}_{t,c} = \sum_{i=0}^{K_t-1}\sum_{j=0}^{K_c-1} \mathbf{W}_{i,j} \cdot \mathbf{X}_{t-i,c-j} + \mathbf{b}, \tag{17}$$

where K_t and K_c represent kernel sizes for temporal and channel dimensions, respectively, enabling simultaneous pattern extraction across time and channels. Examples include TCN and TimesNet.

Transformer-based Models. These models leverage self-attention mechanisms to capture dependencies between temporal and channel elements simultaneously. The attention mechanism first constructs query, key, and value matrices from input:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V, \tag{18}$$

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}, \tag{19}$$

where $\mathbf{X} \in \mathbb{R}^{T\times C}$ can be reshaped to enable cross-attention across both temporal and channel dimensions, allowing flexible relationship modeling. Representative models include Transformer and Informer.

B.2 Interaction Scope Classification

The interaction scope dimension characterizes how models capture dependencies across different temporal positions within each channel. While models built upon the same architecture may vary in the granularity of their temporal interactions modeling, this aspect has often been overlooked in traditional classification frameworks. To address this, we categorize models into four distinct types of interaction scope, as illustrated in Figure 3.

Global Temporal Interaction. Models in this category process all temporal steps within a unified computational framework, enabling long-range dependency modeling. The global processing is formulated as:

$$\mathbf{Y} = \mathcal{F}_{\text{global}}(\mathbf{X}_{1:T}) = \mathcal{F}_{\text{global}}(\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_T), \tag{20}$$

where $\mathcal{F}_{\text{global}}$ represents the model's function that considers all temporal positions simultaneously, enabling unrestricted information flow across the entire sequence. Representative models include RNN, DLinear, Transformer, TSMixer, and PIAD-SRNN.

Local Temporal Interaction. Models focus on interactions within limited temporal windows, utilizing fixed kernel sizes or defined segments. The local processing is formulated as:

$$\mathbf{Y}_t = \mathcal{F}_{\text{local}}(\mathbf{X}_{t-w+1:t}) = \mathcal{F}_{\text{local}}(\mathbf{X}_{t-w+1}, \ldots, \mathbf{X}_t), \tag{21}$$

where w represents the fixed window size, $\mathcal{F}_{\text{local}}$ denotes the local processing function with receptive field limited to w time steps, and boundary conditions are handled through appropriate padding strategies. Representative models include TCN and SegRNN.

Hierarchical Temporal Interaction. Models employing hierarchical structures capture patterns at multiple temporal scales simultaneously. The multi-scale processing combines information from different temporal granularities:

$$\mathbf{Y} = \mathcal{F}_{\text{hierarchical}}\left(\bigoplus_{w \in \mathcal{W}} \alpha_w \cdot \mathcal{F}_w(\mathbf{X}_{t-w+1:t})\right), \tag{22}$$

where $\mathcal{W} = \{w_1, w_2, \ldots, w_n\}$ represents multiple window sizes, $\mathcal{F}_w$ extracts features at scale w, α_w denotes learnable scale-specific weights, $\bigoplus$ represents feature concatenation or weighted combination, and $\mathcal{F}_{\text{hierarchical}}$ hierarchically integrates multi-scale representations. Representative models include Pyraformer and TimeMixer.

Sparse Temporal Interaction. Models selectively focus on important temporal steps rather than processing all temporal interactions uniformly. The sparse selection mechanism can be generally formulated as:

$$\text{Score}(t) = \mathcal{S}_{\text{model}}(\mathbf{X}, t; \boldsymbol{\theta}), \tag{23}$$

$$\mathcal{I} = \{t_1, t_2, \ldots, t_K\} = \arg\text{top-K}_{t \in [1,T]} \text{Score}(t), \tag{24}$$

$$\mathbf{Y} = \mathcal{F}_{\text{sparse}}(\{\mathbf{X}_t \mid t \in \mathcal{I}\}), \tag{25}$$

where $\mathcal{S}_{\text{model}}(\cdot; \boldsymbol{\theta})$ represents model-specific importance scoring functions parameterized by $\boldsymbol{\theta}$ that may utilize attention mechanisms, query-key similarities, or statistical criteria, $\mathcal{I}$ contains the top-K selected temporal positions with $K \ll T$, and $\mathcal{F}_{\text{sparse}}$ processes only the selected time steps. Representative models include Informer and Autoformer.

B.3 Interaction Level Classification

The interaction level dimension describes how models handle relationships between different channels, providing insights into which aspects of the channel information a model focuses on. Depending on the type of information that can be extracted or derived from the raw channel sequence, models can be categorized into three distinct types, as demonstrated in Figure 4.

Direct Channel Interaction. Models utilize raw channels without special preprocessing, directly employing each channel sequence as input. The direct channel interaction is formulated as:

$$\mathbf{Y} = \mathcal{F}_{\text{direct}}(\mathbf{X}) = \mathcal{F}_{\text{direct}}(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_C), \tag{26}$$

where $\mathbf{x}_c \in \mathbb{R}^T$ represents the c-th channel time series, and $\mathcal{F}_{\text{direct}}$ denotes the model's processing function that operates on raw channel sequence without additional transformations. Representative models include RNN and Linear.

Decomposition Channel Interaction. This approach breaks down a time series sequence into multiple interpretable components before processing. Specifically, the decomposition is defined as:

$$\mathbf{X}_{\text{trend}} = \text{MovingAvg}(\mathbf{X}; k), \tag{27}$$

$$\mathbf{X}_{\text{seasonal}} = \text{SeasonalExtract}(\mathbf{X} - \mathbf{X}_{\text{trend}}; p), \tag{28}$$

$$\mathbf{X}_{\text{residual}} = \mathbf{X} - \mathbf{X}_{\text{trend}} - \mathbf{X}_{\text{seasonal}}, \tag{29}$$

$$\mathbf{Y} = \mathcal{F}_{\text{decomp}}(\mathbf{X}_{\text{trend}}, \mathbf{X}_{\text{seasonal}}, \mathbf{X}_{\text{residual}}), \tag{30}$$

where $\text{MovingAvg}(\cdot; k)$ denotes moving average with window size k, $\text{SeasonalExtract}(\cdot; p)$ extracts seasonal patterns with period p, and $\mathcal{F}_{\text{decomp}}$ is a function to process the decomposed components. Representative models include PIAD-SRNN and DLinear.

Spectral Channel Interaction. The frequency domain has been an important perspective that can be easily extracted from a time series. Models following this approach transform the input sequence in a channel from the time domain into the frequency domain. Specifically, a spectral transformation function $\mathcal{T}(\cdot)$ takes a channel sequence as input and returns the complex-valued spectral representations:

$$\mathbf{S}_c = \mathcal{T}(\mathbf{x}_c). \tag{31}$$

The spectral transformation functions include Fast Fourier Transform (FFT), Discrete Fourier Transform (DFT), and Wavelet Transform. Specifically for FFT, the transformation is given by:

$$\mathbf{S}_c[k] = \text{FFT}(\mathbf{x}_c)[k] = \sum_{t=0}^{T-1} \mathbf{x}_c[t] \cdot e^{-2\pi i k t/T}, \tag{32}$$

$$\mathbf{Y} = \text{IFFT}(\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_C) \text{ or } \mathcal{F}_{\text{spectral}}(\mathbf{S}_1, \mathbf{S}_2, \ldots, \mathbf{S}_C). \tag{33}$$

After obtaining the spectral representations for each channel, frequency domain interactions across channels can be modeled using a function $\mathcal{F}_{\text{spectral}}$ or through Inverse Fast Fourier Transformation (IFFT). Representative models include Autoformer and TimesNet.

C EVALUATION RESULTS

C.1 Accuracy Results

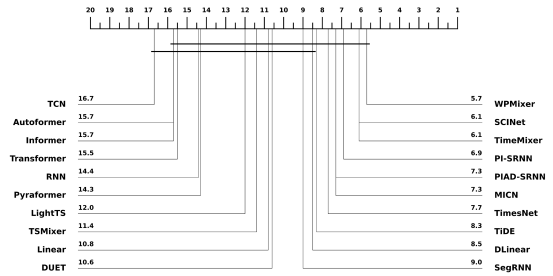
Table 10 reports the same comparison in MAE, which yields the same CD/CI tendencies. Tables 11, and 12 present the entire results for all datasets and models, including evaluations across different prediction horizons.

C.2 Efficiency Results

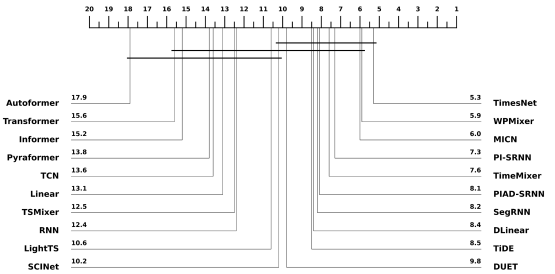
Tables 13 and 14 respectively show the entire inference time and peak GPU memory results of all datasets and models.

C.3 Critical Difference Diagrams

Figures 10 presents comparisons among models using the same training strategy, separating CD and CI approaches. Consistent with the integrated analysis in the main text, these strategy specific comparisons reveal no statistically significant performance differences among models within the same training paradigm across datasets.



(a) Performance comparison based on CD.



(b) Performance comparison based on CI.

Figure 10: Critical difference diagram comparing model rankings under CD and CI strategies across all datasets. Models connected by horizontal bars indicate no statistically significant performance difference.

D DATASET

We use well-established MTSF benchmark datasets [20, 48, 70] spanning multiple domains, including traffic monitoring, economics, and meteorological observations. We further introduce a new dataset, prepared by preprocessing an existing telecommunications dataset [5] tailored for MTSF, to enable a comprehensive assessment across diverse temporal patterns and inter-channel dependencies. As summarized in Table 3, each dataset exhibits distinct channel sizes, lengths, and time granularity, providing comprehensive contexts for effective MTSF evaluation and discussion. All dataset sources are publicly available.

- Traffic [20] contains road occupancy data from San Francisco highways, collected hourly from 2015 to 2016 by 862 road sensors. This dataset represents 24 months of hourly data from the California Department of Transportation, describing road occupancy rates between 0 and 1. The dataset includes complex spatiotemporal patterns of traffic flow, making it an excellent testbed for multivariate time series prediction.
- Weather includes 21 weather indicators (temperature, humidity, wind speed, etc.) collected at 10-minute intervals in Germany during 2020. This dataset reflects complex interdependencies between weather variables, providing insights into meteorological forecasting challenges. The Weather datasets are publicly available at https://www.ncei.noaa.gov/data/local-climatological-data/.
- Electricity Transformer Temperature (ETT) datasets are collected from different regions in China from July 2016 to July 2018. These datasets come in two temporal resolutions: hourly recordings (ETTh1, ETTh2) and 5-minute intervals (ETTm1, ETTm2), with numerical suffixes indicating different geographical locations. Each dataset contains seven variables including oil temperature, load, and power supply measurements, providing diverse patterns of electrical system behavior under varying conditions. The ETT datasets are publicly available at https://github.com/zhouhaoyi/ETDataset.
- Electricity [48] records hourly electricity consumption of 321 customers from 2012 to 2014, with values measured in kilowatts per 15-minute intervals. This dataset serves as an important benchmark for electricity demand forecasting and management.
- Exchange [20] contains daily exchange rates of eight countries (Australia, Britain, Canada, Switzerland, China, Japan, New Zealand, and Singapore) relative to the US dollar, ranging from 1990 to 2016. This long-term financial dataset reflects the irregularity and volatility characteristic of currency markets.
- Weekly Influenza-Like Illness (ILI) contains incidence rate data provided by the US Centers for Disease Control and Prevention (CDC) from 2002 to 2021. The data describes the ratio of patients seen with influenza-like symptoms to the total number of patients, capturing both seasonality and long-term epidemiological trends. The ILI datasets are publicly available at https://gis.cdc.gov/grasp/fluview/fluportaldashboard.html.
- Milano-6165 is a newly constructed dataset derived from the Milano telecommunications dataset [5], which contains mobile phone activity records collected from the Milan region of Italy. It comprises five types of telecommunications activities (SMS-in, SMS-out, calls-in, calls-out, and internet traffic) collected at distinct event times across geographical zones with unique cell IDs. While the original dataset was not explicitly designed for MTSF, it offers diverse communication patterns from various modes, making it well-suited for analyzing inter-channel dependencies. To adapt the original data for MTSF tasks and address issues such as missing values and irrelevant features, we applied the following preprocessing steps to prepare the final dataset presented in Figure 11:
  (1) Removal of unnecessary features (e.g., country code).
  (2) Selection of a single cell ID (6165) with the lowest proportion of missing values among multiple geographical zones.
  (3) Aggregation of event-based records of each activity into real-valued volume within a 10-minute interval, reflecting the intensity of communication behavior over time.
  (4) Setting the internet traffic as the prediction target.

E BRICKTS

E.1 Module Implementations

Algorithms 2, 3, and 4 list the implementation of each option along the Level, Scope, and Architecture axes of BrickTS. Every option takes an input of the same shape and returns a feature of the same shape, so that any combination assembles into a valid model.

E.2 BrickTS Experiments and Results

Tables 15, 16 present comprehensive experimental results for all BrickTS model combinations across all datasets under both CD and CI training strategies. The experiments were conducted using grid search to explore hyperparameters specific to each BrickTS combination, while keeping common hyperparameters fixed across

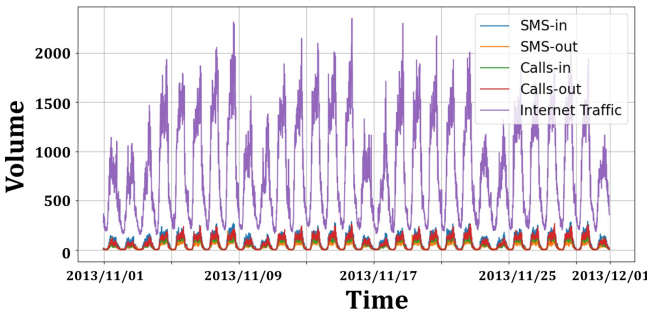


Figure 11: The Milano-6165 dataset with five communication channels (preprocessed from the original dataset [5]).

Algorithm 2 Level Modules: Channel Interaction

Require: Input $\mathbf{X} \in \mathbb{R}^{B\times L\times C}$
Ensure: Feature $\mathbf{Z}_{level} \in \mathbb{R}^{B\times L\times D}$

(a) Direct:
$\mathbf{Z}_{level} \leftarrow \texttt{LayerNorm}(\texttt{Linear}(\mathbf{X}))$

(b) Decomposition:
$\mathbf{X}_{trend} \leftarrow \texttt{MovingAvg}(\mathbf{X}, k)$
$\mathbf{X}_{seasonal} \leftarrow \mathbf{X} - \mathbf{X}_{trend}$
$\mathbf{Z}_t \leftarrow \texttt{Linear}_{trend}(\mathbf{X}_{trend})$
$\mathbf{Z}_s \leftarrow \texttt{Linear}_{seasonal}(\mathbf{X}_{seasonal})$
$\mathbf{Z}_r \leftarrow \texttt{Linear}_{residual}(\mathbf{X})$ ▷ raw input as residual branch
$\mathbf{Z}_{level} \leftarrow \texttt{LayerNorm}(\texttt{Concat}(\mathbf{Z}_t, \mathbf{Z}_s, \mathbf{Z}_r))$

(c) Spectral:
$\mathbf{S} \leftarrow \texttt{rFFT}(\mathbf{X})$ ▷ along time, $\mathbb{R}^{B\times F\times C}$
$\mathbf{S}_K \leftarrow \texttt{TopK}(\mathbf{S}, K)$ ▷ K dominant frequency bins
$\mathbf{Z}_{freq} \leftarrow \texttt{Broad.}(\texttt{Linear}_{freq}(\texttt{Flatten}(\mathbf{S}_K)), L)$
▷ $\texttt{Broad.}(\cdot, L)$ tiles the descriptor along the L time steps
$\mathbf{Z}_{time} \leftarrow \texttt{Linear}_{time}(\mathbf{X})$
$\mathbf{Z}_{level} \leftarrow \texttt{LayerNorm}(\texttt{Linear}(\texttt{Concat}(\mathbf{Z}_{time}, \mathbf{Z}_{freq})))$

Algorithm 3 Scope Modules: Temporal Dependency

Require: Input $\mathbf{X} \in \mathbb{R}^{B\times L\times C}$
Ensure: Feature $\mathbf{Z}_{scope} \in \mathbb{R}^{B\times L\times D}$

(a) Global:
$\mathbf{X}_{mix} \leftarrow \texttt{TokenMixingMLP}(\mathbf{X})$ ▷ mix over time
$\mathbf{Z}_{scope} \leftarrow \texttt{LayerNorm}(\texttt{Linear}(\mathbf{X}_{mix}))$

(b) Local:
$\mathbf{Z}_{scope} \leftarrow \texttt{LayerNorm}(\texttt{GELU}(\texttt{CausalConv1d}(\mathbf{X}, k, d)))$

(c) Hierarchical:
for $k_i \in \{k_1, \ldots, k_n\}$ **do**
  $\mathbf{Z}_i \leftarrow \texttt{Conv1d}(\mathbf{X}, k_i)$
end for
$\boldsymbol{\alpha} \leftarrow \texttt{Softmax}(\mathbf{w})$ ▷ learnable scale weights
$\mathbf{Z}_{cat} \leftarrow \texttt{Concat}(\alpha_1\mathbf{Z}_1, \ldots, \alpha_n\mathbf{Z}_n)$
$\mathbf{Z}_{scope} \leftarrow \texttt{LayerNorm}(\texttt{Linear}(\texttt{GELU}(\mathbf{Z}_{cat})))$

(d) Sparse:
$\mathbf{s} \leftarrow \texttt{ScoreNet}(\mathbf{X})$ ▷ importance scores, $\mathbb{R}^{B\times L}$
$\mathbf{X}_{sel} \leftarrow \texttt{Gather}(\mathbf{X}, \texttt{TopK}(\mathbf{s}, K))$ ▷ K salient steps, $\mathbb{R}^{B\times K\times C}$
$\mathbf{Z}_{sparse} \leftarrow \texttt{MeanPool}(\texttt{Linear}_{sparse}(\mathbf{X}_{sel}))$
$\mathbf{Z}_{scope} \leftarrow \texttt{LayerNorm}(\texttt{Linear}_{full}(\mathbf{X}) + \texttt{Broad.}(\mathbf{Z}_{sparse}, L))$

Algorithm 4 Architecture Modules: Backbone and Prediction

Require: Fused feature $\mathbf{Z} \in \mathbb{R}^{B\times L\times D}$
Ensure: Prediction $\mathbf{Y} \in \mathbb{R}^{B\times H\times C_{out}}$

(a) MLP:
$\mathbf{Y} \leftarrow \texttt{Reshape}(\texttt{MLP}(\texttt{Flatten}(\mathbf{Z})))$

(b) RNN:
$\mathbf{H} \leftarrow \texttt{GRU}(\mathbf{Z})$
$\mathbf{h} \leftarrow \mathbf{H}[:, -1, :]$ ▷ last hidden state
$\mathbf{Y} \leftarrow \texttt{Reshape}(\texttt{Linear}(\mathbf{h}))$

(c) CNN:
for $i = 1$ **to** N **do**
  $\mathbf{Z} \leftarrow \mathbf{Z} + \texttt{GELU}(\texttt{LayerNorm}(\texttt{DilatedConv}(\mathbf{Z}, k, d{=}b^{\,i-1})))$
end for
$\mathbf{Y} \leftarrow \texttt{Reshape}(\texttt{Linear}(\texttt{Flatten}(\mathbf{Z})))$

(d) Transformer:
$\mathbf{Z} \leftarrow \mathbf{Z} + \texttt{PositionalEncoding}$
for $i = 1$ **to** N **do**
  $\mathbf{Z} \leftarrow \texttt{TransformerEncoderLayer}(\mathbf{Z})$
end for
$\mathbf{Y} \leftarrow \texttt{Reshape}(\texttt{Linear}(\texttt{Flatten}(\mathbf{Z})))$

all experiments. The key fixed hyperparameters include a learning rate of 0.001, batch size of 2048, dropout rate of 0.2, model dimension of 128, training epochs of 20, and early stopping patience of 10. Detailed hyperparameter specifications for each model combination can be found in the accompanying code repository.

F REVIN FULL RESULTS

Tables 17 and 18 report the MSE of all 48 BrickTS configurations under CD and CI with RevIN off and on, which underlie the summary in Table 9.

Table 10: Overall performance (MAE) for both CD and CI strategies across models and datasets. Models are categorized by Architecture, Level, and Scope. For each model-dataset pair, the better-performing strategy is highlighted in bold. The last column shows CD/CI win counts and average performance difference (blue: CD better, red: CI better).

Category	Model	Strategy	ETTh1	ETTh2	ETTm1	ETTm2	Exchange	ILI	Electricity	Milano-6165	Traffic	Weather	CD/CI Wins
MLP Direct Global	LightTS	CD	0.2620 (±0.0081)	**0.3980** (±0.0017)	0.1755 (±0.0040)	0.2811 (±0.0010)	**0.3662** (±0.0042)	2.5917 (±0.0735)	0.5335 (±0.0027)	0.3297 (±0.0054)	**0.4302** (±0.0083)	**0.0341** (±0.0024)	4/6 3.42%
		CI	**0.2545** (±0.0075)	0.4657 (±0.0030)	**0.1628** (±0.0086)	**0.2485** (±0.0020)	0.3773 (±0.0094)	**1.2782** (±0.0933)	**0.4330** (±0.0020)	**0.2722** (±0.0059)	0.4868 (±0.0067)	0.0624 (±0.0005)	
MLP Direct Global	Linear	CD	0.2809 (±0.0013)	0.4467 (±0.0030)	0.1786 (±0.0001)	0.2801 (±0.0002)	**0.2544** (±0.0014)	1.4892 (±0.0094)	0.4713 (±0.0011)	**0.3493** (±0.0015)	**0.4271** (±0.0013)	**0.0232** (±0.0001)	4/6 9.93%
		CI	**0.2745** (±0.0018)	**0.4386** (±0.0016)	**0.1589** (±0.0006)	**0.2538** (±0.0018)	1.1490 (±0.0526)	**0.8973** (±0.0083)	**0.4706** (±0.0009)	0.3549 (±0.0017)	0.5407 (±0.0032)	0.0631 (±0.0003)	
MLP Direct Global	TiDE	CD	0.2633 (±0.0002)	**0.4039** (±0.0003)	0.1701 (±0.0000)	0.3030 (±0.0006)	**0.2235** (±0.0009)	0.8829 (±0.0037)	0.4773 (±0.0004)	**0.5912** (±0.0004)	**0.3777** (±0.0008)	**0.0229** (±0.0000)	5/5 1.35%
		CI	**0.2544** (±0.0002)	0.4064 (±0.0003)	**0.1561** (±0.0000)	**0.2495** (±0.0001)	0.2975 (±0.0029)	**0.6969** (±0.0036)	**0.4745** (±0.0007)	0.6966 (±0.0076)	0.4146 (±0.0030)	0.0269 (±0.0002)	
MLP Spectral Sparse	DUET	CD	0.2661 (±0.0003)	**0.4827** (±0.0034)	**0.1681** (±0.0005)	0.2958 (±0.0014)	0.2865 (±0.0013)	0.9239 (±0.0145)	0.4907 (±0.0023)	0.3417 (±0.0016)	**0.4941** (±0.0024)	**0.0249** (±0.0003)	4/6 4.22%
		CI	**0.2575** (±0.0007)	0.4971 (±0.0004)	0.1715 (±0.0001)	**0.2649** (±0.0014)	**0.2375** (±0.0009)	**0.7224** (±0.0060)	**0.4697** (±0.0048)	**0.3376** (±0.0079)	0.5238 (±0.0049)	0.0263 (±0.0001)	
MLP Spectral Hier.	TimeMixer	CD	0.2824 (±0.0017)	**0.4060** (±0.0044)	0.1594 (±0.0002)	0.2710 (±0.0030)	0.2937 (±0.0026)	**0.6530** (±0.0044)	0.4447 (±0.0020)	**0.2635** (±0.0097)	**0.3209** (±0.0054)	0.0272 (±0.0005)	4/6 5.16%
		CI	**0.2802** (±0.0015)	0.4430 (±0.0035)	**0.1591** (±0.0003)	**0.2608** (±0.0014)	**0.2902** (±0.0028)	0.6797 (±0.0401)	**0.4283** (±0.0038)	0.4734 (±0.0788)	0.3457 (±0.0037)	**0.0265** (±0.0001)	
MLP Spectral Hier.	WPMixer	CD	0.2589 (±0.0006)	**0.4036** (±0.0019)	0.1580 (±0.0007)	0.2563 (±0.0012)	0.2815 (±0.0064)	0.8105 (±0.0370)	0.4266 (±0.0015)	**0.4329** (±0.0353)	**0.3124** (±0.0031)	**0.0255** (±0.0004)	4/6 0.78%
		CI	**0.2501** (±0.0006)	0.4563 (±0.0054)	**0.1527** (±0.0002)	**0.2364** (±0.0006)	**0.2373** (±0.0040)	**0.6674** (±0.0091)	**0.3822** (±0.0014)	0.5415 (±0.0011)	0.3563 (±0.0028)	0.0273 (±0.0002)	
MLP Decomp. Global	DLinear	CD	0.2666 (±0.0005)	**0.3873** (±0.0004)	0.1745 (±0.0003)	0.2744 (±0.0002)	**0.2291** (±0.0038)	1.5755 (±0.0079)	0.4548 (±0.0000)	**0.3723** (±0.0002)	**0.3883** (±0.0004)	**0.0232** (±0.0005)	5/5 7.26%
		CI	**0.2485** (±0.0020)	0.3964 (±0.0026)	**0.1523** (±0.0007)	**0.2391** (±0.0009)	0.3091 (±0.0011)	**0.9657** (±0.0066)	**0.4487** (±0.0000)	0.6821 (±0.0011)	0.4244 (±0.0003)	0.0626 (±0.0002)	
MLP Decomp. Global	TSMixer	CD	**0.2654** (±0.0067)	0.4457 (±0.0015)	0.1714 (±0.0015)	0.3265 (±0.0088)	**0.3285** (±0.0466)	1.3160 (±0.0370)	0.5114 (±0.0011)	**0.2984** (±0.0035)	**0.3758** (±0.0078)	**0.0550** (±0.0049)	5/5 3.40%
		CI	0.3024 (±0.0039)	**0.4340** (±0.0056)	**0.1710** (±0.0010)	**0.2677** (±0.0005)	0.3360 (±0.0054)	**1.2506** (±0.0418)	**0.4528** (±0.0068)	0.3592 (±0.0017)	0.4526 (±0.0052)	0.0714 (±0.0008)	
CNN Direct Local	TCN	CD	0.3675 (±0.0352)	0.6050 (±0.0010)	0.2643 (±0.0209)	0.4219 (±0.0122)	1.2844 (±0.3589)	2.7316 (±0.0900)	**0.4676** (±0.0028)	**0.3199** (±0.0249)	0.5253 (±0.0121)	0.0779 (±0.0013)	2/8 21.29%
		CI	**0.3414** (±0.0090)	**0.4176** (±0.0013)	**0.2083** (±0.0182)	**0.2840** (±0.0029)	**0.6434** (±0.0322)	**1.5817** (±0.1279)	0.4678 (±0.0005)	0.3243 (±0.0074)	**0.4670** (±0.0053)	**0.0629** (±0.0027)	
CNN Spectral Hier.	TimesNet	CD	0.2661 (±0.0065)	0.4696 (±0.0290)	0.1627 (±0.0038)	0.2613 (±0.0060)	0.2830 (±0.0074)	0.7384 (±0.0196)	0.4225 (±0.0047)	0.5177 (±0.0140)	**0.2661** (±0.0113)	0.0285 (±0.0006)	1/9 9.84%
		CI	**0.2580** (±0.0062)	**0.4393** (±0.0395)	**0.1532** (±0.0001)	**0.2390** (±0.0029)	**0.2483** (±0.0111)	**0.6661** (±0.0138)	**0.3899** (±0.0050)	**0.2650** (±0.0031)	0.2902 (±0.0116)	**0.0273** (±0.0001)	
CNN Decomp. Hier.	MICN	CD	0.2727 (±0.0050)	0.3928 (±0.0035)	0.1662 (±0.0048)	0.2563 (±0.0003)	**0.2268** (±0.0027)	**0.7957** (±0.0105)	0.5202 (±0.0040)	**0.2984** (±0.0099)	0.3124 (±0.0047)	**0.0427** (±0.0018)	4/6 5.31%
		CI	**0.2535** (±0.0029)	**0.3917** (±0.0031)	**0.1519** (±0.0009)	**0.2276** (±0.0006)	0.3398 (±0.0207)	0.8166 (±0.0047)	**0.3982** (±0.0031)	0.6091 (±0.0455)	**0.2760** (±0.0032)	0.0597 (±0.0029)	
CNN Decomp. Hier.	SCINet	CD	0.2677 (±0.0011)	**0.4194** (±0.0022)	0.1580 (±0.0016)	0.2604 (±0.0026)	0.3469 (±0.0042)	1.1392 (±0.0816)	**0.3975** (±0.0085)	**0.2811** (±0.0035)	**0.2508** (±0.0034)	**0.0260** (±0.0004)	5/5 5.12%
		CI	**0.2670** (±0.0076)	0.4478 (±0.0290)	**0.1553** (±0.0000)	**0.2459** (±0.0008)	**0.2556** (±0.0013)	**0.6807** (±0.0120)	0.4905 (±0.0065)	0.5069 (±0.0176)	0.5124 (±0.0267)	0.0272 (±0.0002)	
RNN Direct Global	RNN	CD	**0.2872** (±0.0019)	**0.4442** (±0.0016)	0.2943 (±0.0113)	0.3232 (±0.0065)	0.4041 (±0.0106)	2.6070 (±0.0170)	0.4590 (±0.0113)	**0.3549** (±0.0092)	0.4959 (±0.0166)	0.0701 (±0.0003)	3/7 13.48%
		CI	0.3057 (±0.0018)	0.4669 (±0.0165)	**0.2211** (±0.0308)	**0.2704** (±0.0029)	**0.3418** (±0.0050)	**0.9393** (±0.1076)	**0.4554** (±0.0025)	0.4221 (±0.0040)	**0.4157** (±0.0062)	**0.0532** (±0.0042)	
RNN Direct Global	PI-SRNN	CD	0.2794 (±0.0010)	0.4346 (±0.0010)	0.1605 (±0.0010)	0.2715 (±0.0005)	**0.2602** (±0.0007)	0.8114 (±0.0110)	0.4591 (±0.0032)	**0.2618** (±0.0041)	**0.3484** (±0.0026)	**0.0249** (±0.0001)	4/6 2.84%
		CI	**0.2598** (±0.0004)	**0.4182** (±0.0006)	**0.1548** (±0.0001)	**0.2390** (±0.0002)	0.2923 (±0.0003)	**0.6704** (±0.0050)	**0.4374** (±0.0017)	0.5111 (±0.0030)	0.3771 (±0.0023)	0.0275 (±0.0001)	
RNN Direct Local	SegRNN	CD	0.2631 (±0.0007)	0.4620 (±0.0019)	0.1571 (±0.0003)	0.2582 (±0.0004)	**0.2205** (±0.0003)	0.9289 (±0.0087)	**0.5085** (±0.0015)	0.3544 (±0.0034)	0.4182 (±0.0010)	0.0382 (±0.0020)	2/8 7.16%
		CI	**0.2599** (±0.0019)	**0.4536** (±0.0005)	**0.1536** (±0.0004)	**0.2380** (±0.0006)	0.2240 (±0.0012)	**0.7075** (±0.0038)	0.5249 (±0.0108)	**0.2949** (±0.0036)	**0.4113** (±0.0026)	**0.0302** (±0.0047)	
RNN Decomp. Global	PIAD-SRNN	CD	**0.2630** (±0.0001)	0.4133 (±0.0002)	0.1732 (±0.0003)	0.2771 (±0.0001)	**0.2374** (±0.0001)	0.8137 (±0.0211)	0.4738 (±0.0003)	**0.3006** (±0.0049)	0.3609 (±0.0016)	**0.0247** (±0.0003)	4/6 3.62%
		CI	0.2724 (±0.0009)	**0.4052** (±0.0003)	**0.1578** (±0.0002)	**0.2494** (±0.0001)	0.3015 (±0.0029)	**0.6527** (±0.0034)	**0.4390** (±0.0003)	0.6104 (±0.0095)	**0.3593** (±0.0013)	0.0272 (±0.0001)	
Trans. Direct Global	Trans.	CD	**0.3148** (±0.0118)	0.5004 (±0.0126)	0.3067 (±0.0462)	0.3684 (±0.0100)	**0.6369** (±0.0852)	1.8638 (±0.0801)	0.5166 (±0.0091)	**0.2348** (±0.0026)	0.5447 (±0.0610)	0.0567 (±0.0166)	3/7 2.80%
		CI	0.3356 (±0.0216)	**0.4568** (±0.0066)	**0.1918** (±0.0223)	**0.3112** (±0.0261)	0.9363 (±0.2681)	**1.7572** (±0.0706)	**0.5135** (±0.0143)	0.4415 (±0.0012)	**0.3871** (±0.0088)	**0.0476** (±0.0030)	
Trans. Direct Sparse	Informer	CD	0.6395 (±0.2008)	0.6891 (±0.0374)	0.2183 (±0.0161)	0.3683 (±0.0257)	0.7412 (±0.0874)	1.9931 (±0.0535)	**0.4911** (±0.0121)	**0.2383** (±0.0026)	0.5998 (±0.0098)	**0.0476** (±0.0087)	3/7 13.87%
		CI	**0.4419** (±0.2764)	**0.4812** (±0.0085)	**0.2059** (±0.0177)	**0.2595** (±0.0013)	**0.4057** (±0.0222)	**1.6649** (±0.0447)	0.5418 (±0.0041)	0.2552 (±0.0076)	**0.5177** (±0.0098)	0.0574 (±0.0075)	
Trans. Spectral Sparse	Autoformer	CD	0.3145 (±0.0068)	**0.5257** (±0.0759)	0.2036 (±0.0140)	0.3469 (±0.0030)	**0.3432** (±0.0384)	**0.7927** (±0.1360)	**0.5934** (±0.0162)	**1.0074** (±0.2546)	**0.3770** (±0.0036)	**0.0921** (±0.0309)	7/3 13.25%
		CI	**0.3103** (±0.0063)	0.5328 (±0.0287)	**0.1913** (±0.0015)	**0.3434** (±0.0006)	0.8218 (±0.1849)	1.3617 (±0.2172)	0.6647 (±0.0470)	1.0325 (±0.0863)	0.4756 (±0.0034)	0.0976 (±0.0274)	
Trans. Decomp. Hier.	Pyraformer	CD	0.4774 (±0.0319)	0.4767 (±0.0128)	0.2697 (±0.0087)	0.3250 (±0.0336)	0.6025 (±0.0621)	1.7948 (±0.0178)	0.4501 (±0.0011)	**0.3003** (±0.0266)	0.4517 (±0.0078)	0.1293 (±0.0243)	1/9 10.58%
		CI	**0.3772** (±0.0411)	**0.4157** (±0.0031)	**0.2346** (±0.0074)	**0.2744** (±0.0030)	**0.4118** (±0.0453)	**1.7851** (±0.0422)	**0.3909** (±0.0056)	0.6323 (±0.0305)	**0.3883** (±0.0131)	**0.0820** (±0.0200)	
Dataset CD/CI Wins			4/16	9/11	1/19	0/20	11/9	3/17	5/15	16/4	12/8	13/7	74/126

Table 11: Performance (MSE) results are shown for both CD and CI strategies across all models and datasets, evaluated over multiple prediction horizon (Part 1/2). Experiments requiring excessive computational resources were omitted and are marked with -. *PH.* denotes prediction horizon.

Dataset	PH.	Autoformer		DLinear		PIAD-SRNN		DUET		Informer		LightTS		Linear		MICN		Pyraformer		RNN	
		CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI
ETTh1	96	0.163	0.154	0.123	0.105	0.115	0.122	0.117	0.112	0.582	0.339	0.118	0.110	0.133	0.124	0.128	0.108	0.321	0.220	0.137	0.149
ETTh2	96	0.445	0.459	0.258	0.268	0.290	0.280	0.378	0.397	0.718	0.378	0.273	0.353	0.325	0.317	0.255	0.261	0.359	0.284	0.309	0.336
ETTm1	96	0.071	0.063	0.060	0.045	0.054	0.047	0.051	0.053	0.083	0.076	0.059	0.050	0.062	0.048	0.055	0.045	0.115	0.091	0.134	0.084
ETTm2	96	0.204	0.204	0.140	0.121	0.145	0.126	0.159	0.133	0.224	0.124	0.143	0.123	0.144	0.129	0.121	0.106	0.177	0.135	0.172	0.134
Electricity	96	0.584	0.716	0.407	0.385	0.436	0.373	0.438	0.409	0.424	0.502	0.502	0.354	0.429	0.412	0.477	0.290	0.347	0.275	0.365	0.385
	192	0.708	0.740	0.379	0.362	0.411	0.372	0.435	0.425	0.551	0.574	0.516	0.360	0.401	0.386	0.460	0.328	0.369	0.292	0.379	0.365
	336	0.695	0.778	0.406	0.388	0.446	0.428	0.477	0.465	0.624	0.649	0.546	0.397	0.427	0.411	0.481	0.374	0.383	0.322	0.445	0.393
Exchange	96	0.196	1.039	0.087	0.141	0.099	0.149	0.136	0.097	0.712	0.243	0.205	0.216	0.101	2.264	0.089	0.179	0.540	0.257	0.239	0.167
	192	0.341	5.958	0.182	0.225	0.213	0.255	0.248	0.203	1.771	0.408	0.304	0.317	0.193	2.031	0.186	0.592	1.367	0.520	0.276	0.287
	336	0.583	11.075	0.333	0.359	0.442	0.450	0.445	0.401	1.515	0.854	0.473	0.515	0.341	2.232	0.302	1.282	1.831	1.065	0.408	0.525
ILI	24	1.256	3.072	3.412	1.169	0.782	0.673	1.105	0.636	5.634	3.532	8.499	2.313	2.749	1.022	1.152	1.332	3.849	3.668	8.029	1.365
	36	1.097	2.718	2.949	1.189	0.945	0.700	1.129	0.748	5.030	3.676	8.549	2.342	2.632	1.074	0.917	0.982	4.192	4.217	7.820	1.176
Milano-6165	24	1.259	1.190	0.217	0.518	0.122	0.439	0.174	0.196	0.142	0.133	0.187	0.114	0.203	0.208	0.178	0.677	0.139	0.473	0.195	0.272
	36	1.605	1.506	0.231	0.618	0.173	0.712	0.230	0.208	0.106	0.119	0.197	0.137	0.213	0.221	0.171	0.599	0.155	0.574	0.223	0.279
	48	1.760	1.653	0.222	0.674	0.209	1.001	0.283	0.250	0.118	0.126	0.204	0.146	0.215	0.215	0.266	0.507	0.171	0.620	0.210	0.270
Traffic	96	0.293	0.428	0.300	0.335	0.281	0.269	0.385	0.441	0.657	0.513	0.359	0.419	0.344	0.508	0.227	0.193	0.421	0.331	0.488	0.332
	192	0.303	0.483	0.261	0.283	0.247	0.245	0.373	0.427	0.805	0.574	0.313	0.379	0.304	0.458	0.233	0.187	0.458	0.356	0.478	0.302
	336	0.333	0.509	0.259	0.275	0.236	0.240	0.380	0.430	0.776	0.651	0.309	0.378	0.303	0.455	0.247	0.202	0.467	0.410	-	0.293
Weather	96	0.013	0.016	0.001	0.006	0.001	0.001	0.001	0.001	0.004	0.006	0.002	0.006	0.001	0.006	0.003	0.006	0.030	0.015	0.008	0.005
	192	0.007	0.022	0.001	0.006	0.001	0.002	0.001	0.001	0.004	0.020	0.003	0.006	0.001	0.006	0.003	0.007	0.024	0.045	0.010	0.005
	336	0.010	0.012	0.001	0.006	0.002	0.002	0.002	0.002	0.006	0.036	0.003	0.006	0.001	0.006	0.005	0.007	0.022	0.008	0.011	0.005

Table 12: Performance(MSE) results are shown for both CD and CI strategies across all models and datasets, evaluated over multiple prediction horizon (Part 2/2). Experiments requiring excessive computational resources were omitted and are marked with -. *PH.* denotes prediction horizon.

Dataset	PH.	SCINet		SegRNN		PI-SRNN		TCN		TiDE		TimeMixer		TimesNet		Transformer		TSMixer		WPMixer	
		CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI	CD	CI
ETTh1	96	0.119	0.120	0.116	0.113	0.127	0.113	0.207	0.194	0.116	0.110	0.130	0.128	0.118	0.112	0.158	0.169	0.119	0.147	0.113	0.107
ETTh2	96	0.300	0.338	0.353	0.344	0.320	0.296	0.548	0.273	0.280	0.281	0.280	0.325	0.360	0.328	0.377	0.334	0.319	0.310	0.278	0.339
ETTm1	96	0.047	0.046	0.047	0.045	0.048	0.046	0.115	0.076	0.053	0.047	0.048	0.048	0.049	0.046	0.150	0.066	0.055	0.054	0.048	0.045
ETTm2	96	0.133	0.125	0.128	0.116	0.140	0.120	0.281	0.142	0.163	0.127	0.139	0.133	0.136	0.124	0.216	0.176	0.174	0.136	0.130	0.118
Electricity	96	0.289	0.444	0.472	0.491	0.410	0.369	0.375	0.408	0.442	0.425	0.352	0.334	0.321	0.287	0.460	0.469	0.467	0.384	0.350	0.296
	192	0.323	0.428	0.491	0.513	0.412	0.375	0.384	0.388	0.415	0.401	0.397	0.385	0.355	0.312	0.517	0.587	0.488	0.388	0.356	0.323
	336	0.389	0.463	0.539	0.603	0.456	0.427	0.397	0.411	-	0.434	0.423	0.453	-	0.384	-	0.675	-	0.426	-	0.374
Exchange	96	0.200	0.115	0.090	0.090	0.115	0.139	2.349	0.596	0.089	0.145	0.143	0.140	0.137	0.112	0.580	1.256	0.161	0.173	0.129	0.099
	192	0.307	0.224	0.196	0.195	0.216	0.250	2.941	0.932	0.194	0.251	0.265	0.256	0.281	0.230	0.779	1.271	0.272	0.469	0.239	0.212
	336	0.503	0.440	0.407	0.391	0.424	0.445	2.837	1.331	0.389	0.443	0.468	0.454	0.455	0.415	1.347	1.886	0.474	0.642	0.434	0.421
ILI	24	1.967	0.699	1.326	0.761	0.763	0.886	8.627	3.472	1.268	1.289	0.632	0.637	0.895	0.839	3.965	3.713	2.621	2.018	0.892	0.731
	36	1.941	0.746	1.243	0.918	0.931	0.833	8.548	3.150	1.042	0.738	0.700	0.734	0.827	0.807	4.329	3.997	2.319	2.249	0.932	0.765
Milano-6165	24	0.114	0.329	0.198	0.124	0.093	0.334	0.157	0.163	0.421	0.571	0.113	0.192	0.323	0.096	0.100	0.230	0.152	0.210	0.248	0.372
	36	0.149	0.523	0.226	0.159	0.135	0.543	0.179	0.185	0.714	0.900	0.123	0.415	0.455	0.125	0.105	0.314	0.157	0.220	0.394	0.599
	48	0.195	0.673	0.242	0.182	0.160	0.709	0.181	0.190	0.973	1.218	0.159	0.496	0.584	0.218	0.103	0.390	0.162	0.212	0.551	0.814
Traffic	96	0.164	0.476	0.337	0.354	0.252	0.287	0.536	0.440	0.292	0.332	0.217	0.259	0.172	0.186	0.570	0.297	0.287	0.383	0.233	0.260
	192	0.174	0.430	0.324	0.342	0.238	0.269	0.515	0.394	-	0.297	0.240	0.229	0.177	0.180	0.657	0.297	0.274	0.356	0.213	0.240
	336	0.178	0.416	0.327	0.336	0.232	0.264	-	0.387	-	0.290	0.244	0.223	0.220	0.175	0.776	0.313	0.284	0.364	0.205	0.234
Weather	96	0.001	0.001	0.002	0.002	0.001	0.001	0.010	0.006	0.001	0.001	0.001	0.001	0.002	0.001	0.005	0.017	0.005	0.007	0.001	0.001
	192	0.002	0.002	0.002	0.002	0.001	0.002	0.011	0.006	0.001	0.002	0.002	0.002	0.002	0.002	0.007	0.003	0.005	0.008	0.001	0.002
	336	0.002	0.002	0.003	0.002	0.002	0.002	0.012	0.006	0.001	0.002	0.002	0.002	0.002	0.002	0.003	0.003	0.005	0.008	0.002	0.002

Table 13: Inference Time (ms) Comparison. Results are shown for both CD and CI strategies across all models and datasets.

Model	Mode	ETTh1	ETTh2	ETTm1	ETTm2	Exchange	ILI	Electricity	Milano-6165	Traffic	Weather
Autoformer	CD	197.31	118.38	683.99	567.40	100.40	3.92	500.57	21.06	207.72	282.49
	CI	185.32	157.62	501.89	589.83	82.99	5.36	721.46	30.10	142.78	511.53
DLinear	CD	101.68	63.48	529.67	205.00	29.64	1.45	420.77	16.54	723.72	66.26
	CI	99.00	43.96	461.97	203.37	19.55	1.45	178.83	16.48	211.56	1.16
PIAD-SRNN	CD	60.74	61.00	131.34	121.23	46.82	0.58	16.33	22.66	27.59	134.73
	CI	76.30	47.70	168.20	201.54	42.87	0.55	93.00	13.21	31.37	329.23
DUET	CD	56.44	53.60	123.74	175.61	37.37	1.57	169.30	46.84	96.96	121.47
	CI	84.45	53.24	161.53	204.50	48.07	2.43	119.40	44.20	24.17	172.30
Informer	CD	160.88	52.67	509.75	456.97	44.71	5.45	375.98	18.10	506.98	249.69
	CI	163.94	45.23	450.25	314.95	36.22	3.28	386.54	18.96	303.20	282.05
LightTS	CD	99.32	52.34	447.37	309.20	39.76	2.69	432.53	9.91	717.68	88.97
	CI	89.90	37.84	468.90	211.17	37.36	2.29	242.37	11.74	221.86	76.57
Linear	CD	83.48	39.43	185.17	131.00	57.17	0.65	81.38	18.45	69.14	283.36
	CI	79.34	43.97	292.63	133.23	31.24	0.79	104.27	19.13	30.12	330.00
MICN	CD	77.01	53.50	420.43	201.39	29.39	3.79	304.23	21.46	785.38	205.97
	CI	111.36	45.14	609.73	226.53	61.43	4.00	323.07	32.67	268.00	333.93
Pyraformer	CD	175.08	108.90	720.07	503.77	48.29	4.69	597.16	10.00	828.58	241.61
	CI	152.11	105.12	722.67	599.97	53.46	3.69	558.33	6.57	505.86	124.24
RNN	CD	78.38	36.33	258.80	106.63	62.23	1.29	169.43	25.14	60.35	264.80
	CI	78.56	41.93	265.00	306.83	51.63	1.33	78.64	26.99	31.31	375.70
SCINet	CD	18.48	23.34	219.46	122.07	31.49	3.94	281.77	23.08	69.42	48.31
	CI	21.89	46.38	90.77	137.46	24.40	4.78	215.80	16.28	10.42	70.15
PI-SRNN	CD	60.67	37.96	75.32	173.03	49.87	1.27	41.83	29.13	30.29	148.41
	CI	66.17	39.80	194.88	110.67	33.60	2.01	101.87	19.57	33.21	396.90
SegRNN	CD	96.70	48.32	441.14	233.73	48.37	2.04	498.83	29.89	55.50	112.20
	CI	87.66	61.22	442.53	230.24	50.23	2.86	226.04	35.53	13.93	178.60
TCN	CD	64.39	69.30	263.90	173.57	56.50	1.85	84.92	12.76	53.52	263.75
	CI	83.06	42.30	205.50	171.00	46.83	1.79	142.28	25.86	33.09	204.67
TSMixer	CD	103.68	30.73	233.13	198.23	25.23	0.57	246.63	14.10	656.42	310.23
	CI	89.12	34.07	259.73	189.53	48.77	1.05	97.91	12.13	176.24	258.80
TiDE	CD	30.43	36.56	3.76	129.17	55.47	1.34	35.61	9.88	67.32	6.54
	CI	66.75	30.94	2.33	89.47	28.56	1.35	101.53	22.68	182.42	322.77
TimeMixer	CD	125.30	35.96	55.74	210.74	13.62	4.68	3.78	50.96	162.64	50.26
	CI	165.80	33.50	19.07	296.80	6.84	4.20	5.40	56.04	54.54	120.12
TimesNet	CD	40.17	59.34	221.31	195.07	61.93	1.05	170.60	14.24	424.43	273.25
	CI	97.66	58.87	231.56	167.23	58.64	1.13	170.28	47.30	264.86	399.57
Transformer	CD	143.68	115.30	732.50	671.71	50.42	2.58	489.80	14.73	721.21	205.96
	CI	190.14	89.18	566.01	368.44	28.37	2.94	507.44	19.03	507.64	147.96
WPMixer	CD	86.96	51.28	325.10	273.07	32.61	0.37	438.40	7.10	688.68	79.40
	CI	123.36	46.56	539.50	338.80	48.00	0.34	282.43	4.03	250.38	61.10

Table 14: Peak GPU Memory (MB) Comparison between CD and CI strategies across all models and datasets.

Model	Mode	ETTh1	ETTh2	ETTm1	ETTm2	Exchange	ILI	Electricity	Milano-6165	Traffic	Weather
Autoformer	CD	91.68	143.58	92.64	91.57	138.35	64.88	128.99	64.60	320.50	93.14
	CI	91.75	142.28	91.72	91.53	477.44	259.76	91.72	63.97	174.86	91.53
DLinear	CD	17.68	17.68	16.84	16.84	16.95	16.52	50.34	16.13	106.56	17.98
	CI	17.85	17.20	17.42	17.42	17.20	16.95	17.42	16.95	17.20	17.20
PIAD-SRNN	CD	18.42	18.42	22.74	18.42	18.58	17.39	66.91	17.45	151.26	20.56
	CI	20.73	17.51	20.73	20.73	19.18	17.22	18.37	17.71	20.80	20.73
DUET	CD	58.91	29.81	21.75	21.78	31.36	20.54	1344.61	19.70	9469.72	27.05
	CI	19.49	19.78	19.79	19.43	38.07	17.77	19.43	17.77	19.36	19.49
Informer	CD	1243.27	620.45	84.23	83.52	68.02	109.27	101.48	133.07	131.27	84.96
	CI	1241.86	83.21	83.91	83.21	67.68	49.05	83.91	132.88	83.91	83.91
LightTS	CD	19.42	24.33	17.92	17.92	17.88	17.39	213.14	19.59	124.17	30.18
	CI	18.09	17.17	18.09	17.22	17.17	17.22	17.22	17.87	17.22	17.22
Linear	CD	20.70	20.70	20.70	16.53	18.59	16.10	38.09	16.22	76.90	17.50
	CI	18.29	18.29	17.09	17.09	17.09	16.99	17.09	16.92	18.29	17.09
MICN	CD	143.25	274.74	274.28	274.74	519.54	159.13	2620.85	496.92	237.47	277.60
	CI	844.12	272.47	272.32	271.44	123.40	159.06	840.20	159.54	141.96	272.32
Pyraformer	CD	299.94	265.61	93.06	93.06	86.71	162.93	197.52	124.73	373.79	99.55
	CI	95.53	85.41	91.11	91.11	258.28	162.34	91.11	224.94	91.11	258.28
RNN	CD	274.23	274.62	190.87	274.62	292.14	61.35	5874.56	84.83	15537.23	529.57
	CI	164.99	164.82	156.14	112.00	164.82	74.73	118.09	74.71	118.36	66.90
SCINet	CD	1.56	1.56	1.56	1.56	5.46	0.99	516.96	0.55	3433.50	5.32
	CI	0.61	0.85	0.61	0.61	0.61	0.64	0.61	0.38	0.61	0.61
PI-SRNN	CD	17.74	21.48	17.93	21.48	21.95	19.07	46.75	17.93	98.06	19.21
	CI	17.19	17.19	17.19	17.42	18.21	19.51	17.42	18.29	17.42	17.42
SegRNN	CD	49.79	49.79	49.79	49.79	239.63	132.11	1104.80	28.31	1783.48	701.83
	CI	30.61	81.26	30.61	30.61	54.34	28.70	30.61	28.70	33.26	81.52
TCN	CD	89.56	203.40	89.56	89.00	97.46	31.20	2863.80	28.69	7659.80	211.74
	CI	55.41	35.69	35.21	34.44	34.44	23.10	35.21	23.10	35.38	148.97
TSMixer	CD	21.29	19.18	32.86	42.81	19.25	23.60	156.12	27.18	89.58	46.97
	CI	41.42	18.76	41.42	41.42	30.10	22.81	18.76	21.81	20.94	41.42
TiDE	CD	32.01	68.43	32.01	230.33	139.34	72.83	119.20	31.95	264.67	35.79
	CI	66.31	30.39	30.39	30.39	41.83	30.64	30.39	23.27	30.39	41.83
TimeMixer	CD	300.49	592.30	592.30	300.49	301.19	43.21	79.23	42.83	424.07	38.60
	CI	296.30	296.30	296.30	296.30	296.30	42.95	36.46	42.65	60.70	156.85
TimesNet	CD	651.70	1775.73	170.97	172.02	2174.36	608.76	191.83	829.49	710.99	171.83
	CI	2202.88	2189.38	171.10	172.72	170.97	451.43	648.75	451.43	170.76	171.10
Transformer	CD	411.85	87.63	102.88	102.88	202.66	53.76	121.31	53.69	151.01	415.27
	CI	106.30	87.34	102.56	102.56	554.11	49.25	102.56	159.93	102.60	102.56
WPMixer	CD	57.65	94.73	29.99	40.52	66.27	132.76	371.59	122.38	955.77	45.48
	CI	25.18	24.81	25.18	23.52	33.52	100.95	23.52	26.31	23.52	24.81

Table 15: Performance (MSE) results of BrickTS with CD and CI training strategies across all datasets and prediction horizons (Part 1/2). CD values are averaged over all channels, whereas CI values are measured on the target channel. Each model is denoted as {Architecture Type}_{Scope Type}_{Level Type} (e.g., rnn_global_decomp.).

Model	Mode	ETTh1	ETTh2	ETTm1	ETTm2	Exchange	ILI	Electricity	Milano-6165	Traffic	Weather
cnn_global_decomp.	CD	0.972	0.561	0.483	0.226	4.577	8.063	0.476	2.095	0.638	0.257
	CI	0.186	0.477	0.086	0.196	2.580	10.477	0.284	1.717	0.326	0.035
cnn_global_direct	CD	0.907	0.479	0.485	0.204	3.954	7.808	0.495	2.183	0.654	0.266
	CI	0.364	0.606	0.102	0.201	1.640	11.994	0.285	2.118	0.346	0.081
cnn_global_spectral	CD	1.299	0.693	0.692	0.269	4.435	8.145	0.543	1.861	0.671	0.397
	CI	1.278	0.877	0.141	0.384	2.034	11.379	0.299	1.832	0.289	0.095
cnn_hier._decomp.	CD	0.901	1.027	0.484	0.211	3.513	7.872	0.458	1.786	0.654	0.283
	CI	0.274	0.602	0.178	0.184	2.105	11.042	0.355	1.920	0.268	0.095
cnn_hier._direct	CD	0.914	0.481	0.492	0.213	3.142	7.870	0.485	1.830	0.654	0.265
	CI	0.313	0.499	0.079	0.227	2.482	10.427	0.360	1.931	0.280	0.019
cnn_hier._spectral	CD	1.077	0.562	0.722	0.257	3.746	8.258	0.529	2.006	0.672	0.320
	CI	0.221	0.801	0.191	0.299	1.334	10.526	0.305	2.090	0.242	0.036
cnn_local_decomp.	CD	0.905	0.473	0.478	0.190	2.723	7.954	0.471	1.843	0.648	0.283
	CI	0.398	0.546	0.067	0.187	1.784	9.905	0.355	1.875	0.274	0.023
cnn_local_direct	CD	0.894	0.559	0.478	0.213	3.390	7.818	0.481	1.967	0.653	0.266
	CI	0.202	0.768	0.098	0.195	4.506	11.728	0.364	1.669	0.281	0.081
cnn_local_spectral	CD	1.216	0.693	0.574	0.217	4.199	7.889	0.584	1.809	0.708	0.332
	CI	0.287	0.653	0.300	0.256	1.716	10.084	0.298	1.669	0.254	0.113
cnn_sparse_decomp.	CD	0.914	0.445	0.473	0.237	2.644	7.981	0.496	1.979	0.654	0.256
	CI	0.417	0.503	0.094	0.191	1.786	9.408	0.330	1.744	0.264	0.061
cnn_sparse_direct	CD	0.875	0.512	0.471	0.253	2.668	7.976	0.509	2.056	0.668	0.270
	CI	0.435	0.528	0.095	0.183	3.671	9.942	0.351	1.684	0.297	0.081
cnn_sparse_spectral	CD	1.432	0.582	0.523	0.235	5.483	8.097	0.679	1.875	0.687	0.306
	CI	0.240	0.496	0.190	0.288	2.754	11.460	0.300	1.859	0.255	0.048
mlp_global_decomp.	CD	1.109	0.555	0.534	0.224	2.258	7.131	0.947	0.704	0.788	0.348
	CI	0.155	0.387	0.056	0.165	0.806	9.347	0.318	0.474	0.223	0.006
mlp_global_direct	CD	1.079	0.485	0.519	0.198	2.144	7.132	1.011	0.759	1.464	0.343
	CI	0.253	0.394	0.064	0.203	1.199	9.491	0.314	0.363	0.211	0.006
mlp_global_spectral	CD	1.197	0.643	0.976	0.266	2.662	7.119	1.007	0.917	1.464	0.597
	CI	0.582	0.474	0.062	0.187	1.602	9.360	0.358	0.648	0.319	0.008
mlp_hier._decomp.	CD	1.034	0.585	0.505	0.219	2.783	7.204	1.005	0.714	1.464	0.358
	CI	0.280	0.384	0.059	0.173	1.486	9.594	0.315	0.340	0.249	0.007
mlp_hier._direct	CD	1.030	0.567	0.548	0.203	2.580	7.127	1.000	0.711	0.989	0.365
	CI	0.758	0.405	0.073	0.176	2.357	9.549	0.305	0.306	0.234	0.007
mlp_hier._spectral	CD	1.150	0.651	0.554	0.233	2.781	7.103	1.012	0.747	1.260	0.420
	CI	0.297	0.543	0.068	0.168	1.475	9.435	0.315	0.356	0.193	0.007
mlp_local_decomp.	CD	1.011	0.560	0.520	0.225	2.297	7.145	1.007	0.704	1.464	0.322
	CI	0.350	0.368	0.057	0.160	2.213	9.440	0.315	0.332	0.203	0.007
mlp_local_direct	CD	1.046	0.546	0.532	0.225	2.554	7.153	1.004	0.712	1.239	0.311
	CI	0.294	0.377	0.061	0.178	1.735	9.357	0.309	0.318	0.212	0.007
mlp_local_spectral	CD	1.057	0.638	0.546	0.215	2.320	7.148	1.013	0.727	1.464	0.425
	CI	0.613	0.421	0.084	0.180	1.830	9.363	0.327	0.347	0.197	0.007
mlp_sparse_decomp.	CD	0.992	0.621	0.494	0.238	2.303	7.176	1.009	0.697	1.464	0.375
	CI	0.434	0.406	0.057	0.190	1.594	9.391	0.325	0.333	0.239	0.007
mlp_sparse_direct	CD	1.026	0.559	0.507	0.263	2.125	7.176	0.951	0.663	1.464	0.310
	CI	0.650	0.387	0.054	0.147	1.626	9.600	0.303	0.371	0.213	0.006
mlp_sparse_spectral	CD	1.151	0.539	0.558	0.217	2.438	7.200	1.013	0.731	1.464	0.625
	CI	0.371	0.431	0.080	0.152	1.505	9.430	0.319	0.337	0.198	0.007

Table 16: Performance (MSE) results of BrickTS with CD and CI training strategies across all datasets and prediction horizons (Part 2/2). CD values are averaged over all channels, whereas CI values are measured on the target channel. Each model is denoted as {Architecture Type}_{Scope Type}_{Level Type} (e.g., rnn_global_decomp.).

Model	Mode	ETTh1	ETTh2	ETTm1	ETTm2	Exchange	ILI	Electricity	Milano-6165	Traffic	Weather
rnn_global_decomp.	CD	1.047	0.536	0.566	0.222	2.579	7.166	0.533	1.040	0.673	0.227
	CI	0.192	0.404	0.054	0.208	1.974	9.450	0.345	0.708	0.277	0.005
rnn_global_direct	CD	1.033	0.514	0.566	0.228	2.446	7.119	0.534	1.019	0.670	0.238
	CI	0.240	0.410	0.052	0.177	1.921	9.568	0.334	0.704	0.270	0.006
rnn_global_spectral	CD	1.121	0.660	0.657	0.221	2.745	7.188	0.558	1.088	0.689	0.256
	CI	0.157	0.402	0.059	0.207	2.477	9.423	0.376	0.807	0.260	0.007
rnn_hier._decomp.	CD	1.047	0.501	0.606	0.222	2.423	7.152	0.534	1.017	0.720	0.283
	CI	0.184	0.418	0.051	0.222	1.681	9.421	0.392	0.647	0.321	0.006
rnn_hier._direct	CD	1.035	0.472	0.602	0.230	2.420	7.158	0.537	1.020	0.731	0.288
	CI	0.193	0.418	0.051	0.218	1.933	9.611	0.401	0.677	0.294	0.006
rnn_hier._spectral	CD	1.045	0.499	0.608	0.232	2.429	7.202	0.550	0.996	0.687	0.271
	CI	0.164	0.397	0.053	0.212	1.838	9.537	0.395	0.661	0.273	0.005
rnn_local_decomp.	CD	1.030	0.487	0.603	0.218	2.539	7.213	0.535	1.005	0.705	0.279
	CI	0.190	0.421	0.052	0.217	1.839	9.814	0.457	0.662	0.324	0.006
rnn_local_direct	CD	1.039	0.522	0.609	0.210	2.525	7.165	0.538	1.038	0.735	0.299
	CI	0.225	0.431	0.053	0.213	1.708	9.356	0.382	0.652	0.345	0.006
rnn_local_spectral	CD	1.051	0.494	0.619	0.227	2.457	7.192	0.547	1.023	0.682	0.267
	CI	0.169	0.397	0.054	0.213	1.856	9.436	0.348	0.660	0.271	0.006
rnn_sparse_decomp.	CD	1.021	0.479	0.609	0.220	2.459	7.157	0.520	1.020	0.710	0.241
	CI	0.180	0.415	0.051	0.214	1.927	9.496	0.386	0.675	0.293	0.006
rnn_sparse_direct	CD	1.022	0.491	0.580	0.209	2.459	7.203	0.540	1.005	0.694	0.256
	CI	0.200	0.426	0.052	0.219	1.795	9.428	0.338	0.624	0.293	0.006
rnn_sparse_spectral	CD	1.079	0.509	0.591	0.215	2.575	7.209	0.539	1.006	0.681	0.256
	CI	0.162	0.403	0.058	0.196	1.909	9.381	0.379	0.640	0.267	0.005
trans._global_decomp.	CD	1.004	0.518	0.447	0.371	1.093	7.600	0.385	1.143	0.646	0.238
	CI	0.224	0.605	0.112	0.223	0.668	10.993	0.297	1.062	0.275	0.030
trans._global_direct	CD	0.681	0.401	0.537	0.239	1.335	7.377	0.408	1.105	0.645	0.221
	CI	0.240	0.654	0.202	0.194	0.746	10.711	0.298	1.071	0.271	0.040
trans._global_spectral	CD	1.086	0.727	0.492	0.184	2.811	7.465	0.555	1.115	0.643	0.316
	CI	0.283	0.535	0.085	0.463	2.464	10.295	0.289	1.122	0.218	0.042
trans._hier._decomp.	CD	0.751	0.510	0.506	0.159	1.256	7.906	0.535	0.813	0.648	0.298
	CI	0.218	0.453	0.170	0.285	1.038	10.054	0.305	0.917	0.263	0.076
trans._hier._direct	CD	0.668	0.419	0.488	0.384	1.429	7.711	0.466	0.894	0.637	0.360
	CI	0.324	0.610	0.112	0.174	0.863	9.422	0.282	1.065	0.274	0.038
trans._hier._spectral	CD	0.894	0.724	0.478	0.178	1.607	7.856	0.462	0.885	0.640	0.256
	CI	0.259	0.833	0.099	0.187	1.060	10.455	0.293	0.850	0.240	0.091
trans._local_decomp.	CD	0.669	0.435	0.601	0.247	1.179	7.468	0.541	0.966	0.640	0.322
	CI	0.339	0.544	0.181	0.368	1.001	10.012	0.289	0.871	0.269	0.028
trans._local_direct	CD	0.760	0.485	0.572	0.238	1.482	7.796	0.471	1.040	0.651	0.262
	CI	0.251	0.458	0.110	0.179	0.826	11.155	0.287	0.745	0.252	0.049
trans._local_spectral	CD	0.792	0.412	0.506	0.174	1.337	8.062	0.489	0.825	0.651	0.255
	CI	1.075	0.595	0.105	0.263	1.408	10.645	0.300	0.791	0.237	0.048
trans._sparse_decomp.	CD	0.783	0.419	0.526	0.249	1.278	7.543	0.411	0.923	0.659	0.258
	CI	0.594	1.023	0.093	0.289	1.234	11.092	0.302	0.782	0.255	0.049
trans._sparse_direct	CD	0.721	0.488	0.684	0.273	1.030	7.509	0.554	0.821	0.664	0.337
	CI	0.399	0.462	0.204	0.246	0.921	9.951	0.304	0.779	0.266	0.058
trans._sparse_spectral	CD	0.764	0.378	0.507	0.191	0.949	7.858	0.529	0.832	0.666	0.236
	CI	0.224	0.476	0.179	0.204	1.419	9.376	0.300	0.806	0.224	0.056

Table 17: Performance (MSE) results of BrickTS with CD and CI training strategies without (w/o) and with (w/) RevIN (Part 1/2). All values are measured on the target channel. Each model is denoted as {Architecture Type}_{Scope Type}_{Level Type} (e.g., rnn_global_decomp.).

		ETTh1		ILI		Electricity		Milano-6165		Weather	
Model	**Mode**	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/
cnn_global_decomp.	CD	0.879	0.133	11.437	1.385	0.616	0.634	2.409	1.101	0.034	0.001
	CI	0.186	0.130	10.477	1.548	0.284	0.289	4.001	1.151	0.035	0.001
cnn_global_direct	CD	0.662	0.136	11.000	1.367	0.697	0.641	2.781	1.091	0.046	0.001
	CI	0.364	0.126	11.994	1.645	0.285	0.293	2.926	1.316	0.081	0.001
cnn_global_spectral	CD	1.348	0.138	10.547	1.539	0.635	0.705	1.248	1.264	0.119	0.002
	CI	1.278	0.140	11.379	1.575	0.299	0.293	2.061	1.247	0.095	0.001
cnn_hier._decomp.	CD	0.625	0.132	10.751	1.665	0.617	0.617	3.474	1.090	0.046	0.001
	CI	0.274	0.122	11.042	1.634	0.355	0.290	2.476	1.144	0.095	0.002
cnn_hier._direct	CD	0.765	0.133	10.171	1.780	0.660	0.603	2.716	1.086	0.034	0.001
	CI	0.313	0.121	10.427	1.666	0.360	0.322	4.017	1.130	0.019	0.001
cnn_hier._spectral	CD	1.180	0.142	10.701	1.902	0.719	0.804	1.684	1.080	0.056	0.001
	CI	0.221	0.133	10.526	1.455	0.305	0.306	3.591	0.995	0.036	0.001
cnn_local_decomp.	CD	0.573	0.134	9.984	1.793	0.645	0.603	4.689	1.100	0.040	0.002
	CI	0.398	0.125	9.905	1.583	0.355	0.292	3.864	1.391	0.023	0.001
cnn_local_direct	CD	0.765	0.133	9.586	1.660	0.662	0.603	4.022	1.149	0.046	0.001
	CI	0.202	0.121	11.728	1.558	0.364	0.316	4.303	1.244	0.081	0.002
cnn_local_spectral	CD	1.150	0.143	9.742	1.599	0.766	0.861	1.884	1.054	0.108	0.002
	CI	0.287	0.132	10.084	1.482	0.298	0.304	4.286	1.086	0.113	0.001
cnn_sparse_decomp.	CD	0.628	0.132	11.267	1.735	0.637	0.611	2.504	1.204	0.036	0.001
	CI	0.417	0.127	9.408	1.513	0.330	0.292	3.536	1.096	0.061	0.001
cnn_sparse_direct	CD	0.658	0.133	11.524	1.667	0.715	0.609	3.023	1.097	0.036	0.002
	CI	0.435	0.121	9.942	1.388	0.351	0.318	4.654	1.150	0.081	0.002
cnn_sparse_spectral	CD	1.325	0.132	10.860	1.416	0.836	0.720	2.340	1.006	0.066	0.001
	CI	0.240	0.135	11.460	1.829	0.300	0.298	2.411	1.302	0.048	0.001
mlp_global_decomp.	CD	1.015	0.138	9.560	1.115	1.000	0.961	0.429	0.821	0.010	0.002
	CI	0.155	0.118	9.347	1.101	0.318	0.310	0.436	0.930	0.006	0.001
mlp_global_direct	CD	1.011	0.138	9.373	1.116	1.050	0.935	0.430	0.813	0.012	0.001
	CI	0.253	0.120	9.491	1.097	0.314	0.304	0.409	0.936	0.006	0.001
mlp_global_spectral	CD	1.238	0.138	9.410	1.122	1.047	0.938	0.600	1.117	0.007	0.002
	CI	0.582	0.131	9.360	1.110	0.358	0.325	0.630	1.197	0.008	0.001
mlp_hier._decomp.	CD	0.916	0.137	9.619	1.118	1.045	0.899	0.423	0.703	0.012	0.001
	CI	0.280	0.111	9.594	1.130	0.315	0.290	0.495	0.812	0.007	0.001
mlp_hier._direct	CD	0.735	0.136	9.414	1.130	1.040	0.924	0.496	0.697	0.010	0.001
	CI	0.758	0.112	9.549	1.127	0.305	0.298	0.458	0.786	0.007	0.001
mlp_hier._spectral	CD	0.991	0.139	9.278	1.074	1.050	0.924	0.507	0.818	0.009	0.001
	CI	0.297	0.115	9.435	1.124	0.315	0.329	0.484	0.771	0.007	0.001
mlp_local_decomp.	CD	0.802	0.135	9.521	1.123	1.047	0.958	0.464	0.718	0.011	0.001
	CI	0.350	0.111	9.440	1.106	0.315	0.304	0.547	0.813	0.007	0.001
mlp_local_direct	CD	0.962	0.136	9.513	1.119	1.042	0.957	0.471	0.739	0.012	0.001
	CI	0.294	0.112	9.357	1.111	0.309	0.297	0.479	0.797	0.007	0.001
mlp_local_spectral	CD	0.990	0.138	9.534	1.119	1.051	0.964	0.482	0.701	0.011	0.002
	CI	0.613	0.115	9.363	1.098	0.327	0.313	0.440	0.775	0.007	0.001
mlp_sparse_decomp.	CD	0.867	0.135	9.492	1.119	1.048	0.959	0.499	0.790	0.009	0.001
	CI	0.434	0.117	9.391	1.100	0.325	0.310	0.377	0.900	0.007	0.001
mlp_sparse_direct	CD	0.775	0.135	9.589	1.108	1.000	0.961	0.439	0.781	0.012	0.002
	CI	0.650	0.115	9.600	1.120	0.303	0.297	0.519	0.885	0.006	0.001
mlp_sparse_spectral	CD	1.146	0.138	9.490	1.098	1.051	0.966	0.430	0.798	0.007	0.001
	CI	0.371	0.117	9.430	1.104	0.319	0.325	0.409	0.866	0.007	0.001

Table 18: Performance (MSE) results of BrickTS with CD and CI training strategies without (w/o) and with (w/) RevIN (Part 2/2). All values are measured on the target channel. Each model is denoted as {Architecture Type}_{Scope Type}_{Level Type} (e.g., rnn_global_decomp.).

		ETTh1		ILI		Electricity		Milano-6165		Weather	
Model	**Mode**	w/o	w/	w/o	w/	w/o	w/	w/o	w/	w/o	w/
rnn_global_decomp.	CD	0.711	0.133	9.457	1.132	0.689	0.600	0.611	1.184	0.012	0.001
	CI	0.192	0.113	9.450	1.134	0.345	0.351	0.558	1.035	0.005	0.001
rnn_global_direct	CD	0.630	0.134	9.303	1.125	0.686	0.601	0.610	1.224	0.012	0.001
	CI	0.240	0.115	9.568	1.132	0.334	0.335	0.558	1.018	0.006	0.001
rnn_global_spectral	CD	0.809	0.137	9.659	1.119	0.722	0.612	0.780	1.245	0.010	0.001
	CI	0.157	0.115	9.423	1.112	0.376	0.367	0.607	1.033	0.007	0.001
rnn_hier._decomp.	CD	0.595	0.133	9.493	1.084	0.677	0.599	0.614	1.199	0.011	0.001
	CI	0.184	0.115	9.421	1.111	0.392	0.385	0.550	1.035	0.006	0.001
rnn_hier._direct	CD	0.670	0.132	9.336	1.082	0.680	0.602	0.602	1.189	0.012	0.001
	CI	0.193	0.115	9.611	1.104	0.401	0.422	0.556	1.035	0.006	0.001
rnn_hier._spectral	CD	0.631	0.135	9.756	1.108	0.710	0.600	0.617	1.180	0.012	0.001
	CI	0.164	0.113	9.537	1.113	0.395	0.489	0.526	1.005	0.005	0.001
rnn_local_decomp.	CD	0.643	0.131	9.552	1.117	0.674	0.597	0.627	1.220	0.013	0.001
	CI	0.190	0.115	9.814	1.108	0.457	0.381	0.560	1.037	0.006	0.001
rnn_local_direct	CD	0.632	0.132	9.459	1.115	0.697	0.600	0.638	1.211	0.012	0.001
	CI	0.225	0.116	9.356	1.102	0.382	0.385	0.554	1.056	0.006	0.001
rnn_local_spectral	CD	0.684	0.132	9.439	1.104	0.712	0.596	0.643	1.166	0.011	0.001
	CI	0.169	0.113	9.436	1.116	0.348	0.419	0.523	1.010	0.006	0.001
rnn_sparse_decomp.	CD	0.618	0.131	9.561	1.093	0.621	0.597	0.596	1.182	0.010	0.001
	CI	0.180	0.116	9.496	1.103	0.386	0.358	0.522	1.030	0.006	0.001
rnn_sparse_direct	CD	0.704	0.132	9.613	1.087	0.688	0.598	0.577	1.149	0.012	0.001
	CI	0.200	0.116	9.428	1.091	0.338	0.394	0.515	1.021	0.006	0.001
rnn_sparse_spectral	CD	0.703	0.132	9.670	1.114	0.656	0.600	0.562	1.161	0.010	0.001
	CI	0.162	0.114	9.381	1.106	0.379	0.413	0.469	1.027	0.005	0.001
trans._global_decomp.	CD	0.571	0.131	10.908	1.405	0.532	0.554	1.420	1.026	0.049	0.002
	CI	0.224	0.131	10.993	1.357	0.297	0.338	1.233	1.187	0.030	0.001
trans._global_direct	CD	0.254	0.127	10.350	1.475	0.613	0.583	1.241	1.102	0.039	0.002
	CI	0.240	0.117	10.711	1.369	0.298	0.329	1.547	1.137	0.040	0.001
trans._global_spectral	CD	0.714	0.137	10.100	1.460	0.704	0.940	2.611	1.399	0.069	0.001
	CI	0.283	0.134	10.295	1.084	0.289	0.303	1.918	1.346	0.042	0.001
trans._hier._decomp.	CD	0.320	0.121	10.048	1.265	0.738	0.682	0.788	1.044	0.127	0.002
	CI	0.218	0.134	10.054	1.194	0.305	0.344	1.000	1.028	0.076	0.002
trans._hier._direct	CD	0.268	0.123	9.705	1.315	0.625	0.636	1.266	1.053	0.207	0.002
	CI	0.324	0.128	9.422	1.202	0.282	0.303	0.967	1.052	0.038	0.001
trans._hier._spectral	CD	0.394	0.128	11.219	1.325	0.586	0.645	1.754	1.053	0.034	0.001
	CI	0.259	0.116	10.455	1.483	0.293	0.320	1.673	0.929	0.091	0.001
trans._local_decomp.	CD	0.260	0.118	10.217	1.431	0.710	0.680	0.688	0.939	0.141	0.002
	CI	0.339	0.120	10.012	1.492	0.289	0.337	1.178	0.975	0.028	0.002
trans._local_direct	CD	0.319	0.118	9.728	1.444	0.649	0.758	0.768	0.946	0.063	0.002
	CI	0.251	0.121	11.155	1.465	0.287	0.318	0.707	0.992	0.049	0.001
trans._local_spectral	CD	0.377	0.126	12.357	1.348	0.629	0.595	1.350	1.094	0.039	0.002
	CI	1.075	0.125	10.645	1.438	0.300	0.322	1.182	1.102	0.048	0.001
trans._sparse_decomp.	CD	0.367	0.124	9.304	1.326	0.499	0.588	0.685	1.030	0.061	0.002
	CI	0.594	0.132	11.092	1.385	0.302	0.334	0.911	1.072	0.049	0.001
trans._sparse_direct	CD	0.389	0.121	9.198	1.403	0.716	0.612	1.091	1.089	0.163	0.002
	CI	0.399	0.125	9.951	1.438	0.304	0.304	1.132	1.041	0.058	0.001
trans._sparse_spectral	CD	0.278	0.136	11.153	1.274	0.694	0.603	1.260	1.148	0.036	0.002
	CI	0.224	0.120	9.376	1.324	0.300	0.308	1.410	1.176	0.056	0.001